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# Ethnic variation in lower face anthropometry on facial computed tomography scans for patients seeking facial feminization surgery

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**Summary** *Background:* Facial feminization surgery (FFS) is the most common form of facial gender-affirming surgery. One of the current knowledge gaps is the understanding of differences among racial groups in baseline craniofacial norms for transgender and nonbinary patients.
*Methods:* All patients who sought consultation for FFS and underwent craniofacial computed tomography (CT) scans at a single institution between 2018 and 2023 were included. Patients who underwent previous facial surgeries were excluded. Chart reviews were conducted for patient characteristics, including race, age, hormone therapy duration, and prior gender-affirming surgeries. Racial categorizations included White, Latinx, African American, or Asian. Patients with other or multiracial identities were excluded. Lower face measurements were derived from preoperative facial CT scans. Comparative analyses were performed on all measurements among the racial groups.
*Results:* In this study, 204 patients were included with an average age of 32.0 ± 10.2 years and a median hormone therapy duration of 2.0 years. The notable differences among the racial groups were: 1. Zygomatic width was the largest in Asian patients (13.5 ± 0.6 cm) compared to all other racial groups ($p = 0.03$), 2. Nasolabial angle was the smallest in African American patients (82.5 ± 13.1 degrees, $p < 0.001$), 3. Lower face height was the largest in African

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American patients (6.9 ± 0.7 cm, $p < 0.001$), and 4. Lateral mandibular flare was the largest in African American patients (0.4 ± 0.1 cm) and the smallest in Latinx patients (0.2 ± 0.1 cm, $p < 0.001$).

*Conclusions:* Specific target areas of FFS should be carefully considered to account for possible baseline ethnic differences. Relative facial proportions may also be a more salient surgical planning tool in transgender and gender nonbinary patients rather than absolute measurements alone.

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Incongruence of the facial features with the experienced gender in transgender and gender nonbinary (TGNB) individuals may lead to a constant dissonance in self-perceptions of identity and, thus, are a common anatomic source for gender dysphoria.[1] Particularly in TGNB individuals assigned male at birth, facial characteristics have been reported to constitute a disproportionate and significant source of dysphoria compared to TGNB individuals assigned female at birth.[2] Thus, most TGNB patients seeking facial reconstruction also seek feminization procedures.

Although there has been a substantial increase in FFS literature in the past decade, a notable gap exists in the elucidation of racial and ethnic variations in underlying craniofacial norms and their relevance when approaching surgical planning and execution in patients undergoing gender-affirming procedures. All anthropometric data in the literature have been derived from cisgender as opposed to transgender individuals, whose physiology may be affected by long-term hormone therapy.

Most anthropometric studies also obtain measurements from photographs or direct calipers with large variations in patient positioning, which may limit consistency and accuracy.[3,4] Owing to the limited nature of two-dimensional photographs, variations between the natural head position and Frankfurt plane parallel to the ground may lead to measurement limitations when compared to a facial computed tomography (CT) scan. Measurements obtained from direct calipers also differ based on the pressure applied by the measuring instrument. Furthermore, measurements from photographs and direct calipers cannot account for osseous versus soft tissue contributions to the overall face shape, which is essential for surgical planning purposes. In this study, we detail the baseline racial variations in the lower face using facial CT scans of TGNB individuals seeking FFS with the objective of enhancing patient-centered preoperative planning.

## Methods

### Patient selection

All patients who sought consultation for FFS and underwent preoperative facial CT scans between 2018 and 2023 at a single institution were included for retrospective analysis. Patients with previous facial surgeries were excluded. Chart reviews were conducted for patient characteristics, including race, age at the time of CT scan, years on hormone therapy, gender identity, and prior gender-affirming surgeries. Racial categorizations included White, Latinx, African American, or Asian. Patients who identified as Middle Eastern, Indigenous, and/or multiracial were excluded. Institutional review board approval was obtained (IRB #22001299).

### Measurements

Precise measurements of the lower face were obtained from manual review of preoperative facial CT scans. The measurements included nasolabial angle, nasofacial angle, osseous upper facial height, osseous lower facial height, soft tissue lower face height, zygomatic width, mandible width, mandible length, ramus length, mandibular angle, and lateral mandibular flare (Figure 1). For mandible length, ramus length, mandibular angle, and lateral mandibular flare, the average of left and right sided measurements were calculated for the final analysis. Other derived calculations included total facial height (sum of upper and lower facial height), ratio of upper to lower facial height (UFH/LFH), mandible width to total facial height, zygomatic width to total facial height, zygoma width to mandible width, and ramus height to mandible length. Details for each measurement are provided in Supplementary Digital Content, Tables 1 and 2. All measurements were carried out once by one investigator and also a second time by one of three other investigators who were blinded to each other's measurements and patient demographic details. To assess the accuracy of measurements, an inter-user reliability coefficient was obtained with the resultant Cronbach's alpha greater than 0.86 for all measurements. Averages between two user measurements were calculated for all final results of the study.

### Statistical analysis

Descriptive statistics were used to evaluate patient clinical and surgical characteristics. The Kolmogorov-Smirnov and Shapiro-Wilk tests were used to assess the normal distribution of quantitative data and the Levene's test was used to assess the homogeneity of variances. Differences in facial measurements among races were compared using one-way analysis of variance (ANOVA) and Kruskal-Wallis tests as appropriate. Specific comparisons between racial groups were analyzed using the Tukey post hoc test and Bonferroni-corrected pairwise comparisons when relevant.

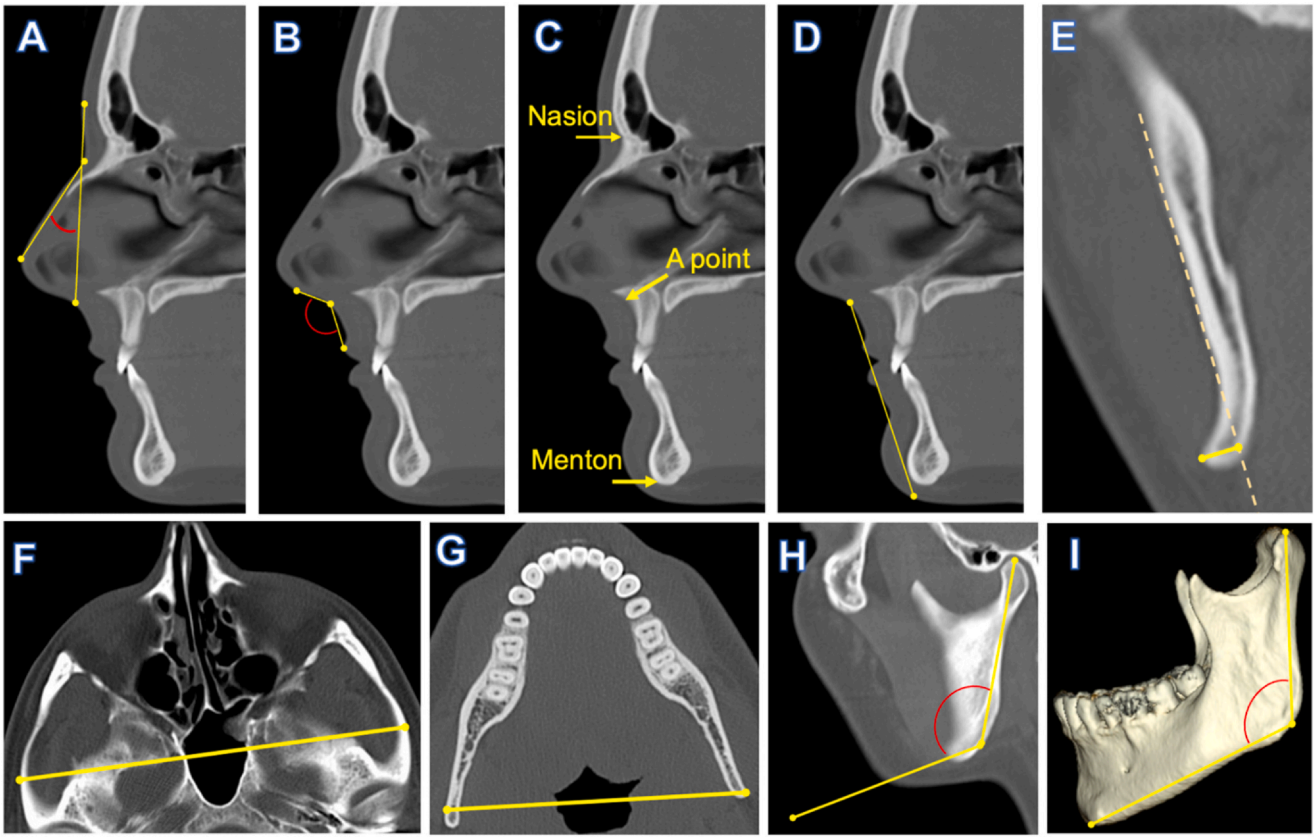


**Figure 1** Absolute lower face measurements and landmarks on CT scans. A) Nasofrontal angle. B) Nasolabial angle. C) Landmarks nasion, A point, and menton used for upper facial height and lower facial height. D) Soft tissue lower face height. E) Lateral mandibular flare was measured by drawing a tangent along the lateral cortex of the ramus in the coronal plane, then measuring the lateral projection from this line perpendicularly. F) Zygomatic width. G) Mandible width. H) Mandible length, ramus length, and mandibular angle. I) 3D representation of mandible length, ramus length, and mandibular angle. Please see Supplementary Digital Content, Tables 1 and 2 for full details on measurements and landmarks.

To ensure the persistence of racial differences within the influence of other potentially impactful variables, linear regression models were constructed for all measurements that exhibited racial differences in ANOVA comparisons. Selected variables in the regression models were clinically relevant to craniofacial size, including body mass index (BMI), age, and duration of hormone replacement therapy (HRT), in addition to White, Latinx, African American, and Asian races as the binary categorical variables. All of the following assumptions were met for each model: linearity and homoscedasticity using a residual versus predicted values plot, absence of collinearity using a variance inflation factor $< 10$, independence of residuals using a Durbin Watson criteria of 1.5–2.5, and normality of residuals using a visual inspection of the P-P plot. All statistical analyses were performed using SPSS, Version 27.0 (SPSS Inc. Chicago, IL) with a significance threshold of $p < 0.05$.

## Results

### Patient characteristics

A total of 213 patients with known racial identities and completed facial CT scans were identified. Nine patients who identified as Middle Eastern, Indigenous, or multiracial were excluded from the analysis. The final study comprised 204 patients (mean age 32.0 ± 10.2 years; Table 1). The most common racial group was White (60.3 %) with Latinx being the second most common (24.0 %). Most patients were on hormone therapy at the time of the CT scan (97.7 %) with a median duration of 2.0 (IQR: 1.0–4.0) years.

Seven out of twelve absolute lower face measurements revealed significant differences between at least two racial groups, including zygoma width, nasolabial angle, lower facial height, total facial height, soft tissue lower face height, mandible length, and lateral mandibular flare (Table 2). Three out of five facial ratios showed significant differences, including UFH/LFH, mandible width to total facial height, and zygoma width to total facial height (Table 3). Inter-racial variations in absolute and relative lower face measurements are summarized in Figures 2 and 3, whereas Figure 4 shows a visual depiction of the average lower face measurements that were significantly different for each racial group.

### Absolute axial measurements

Among the absolute axial measurements, we found differences in ethnicity for only zygomatic width (Table 2). Asian patients had the largest zygomatic width (13.5 ± 0.6 cm)

**Table 1** Patient characteristics.

| Variables | |
|---|---|
| N | 204 |
| Age at CT scan, yrs, mean ± SD | 32.0 ± 10.2 |
| Gender, n (%) | |
| Binary | 173 (84.8) |
| Non-binary | 16 (7.8) |
| Unknown | 15 (7.4) |
| Previous gender-affirming surgeries, n (%) | 109 (53.4) |
| Top surgery | 61 (29.9) |
| Bottom surgery | 49 (24.0) |
| Race, n (%) | |
| White | 123 (60.3) |
| Latinx | 49 (24.0) |
| African American | 12 (5.9) |
| Asian | 20 (9.8) |
| Hormone therapy, yrs, median (IQR) | 2.0 (1.0-4.0) |

compared with White (13.1 ± 0.5 cm, p = 0.003), Latinx (13.2 ± 0.5 cm, p = 0.05), and African American (13.0 ± 0.4 cm, p = 0.02) patients (Figure 2).

## Absolute sagittal measurements

Among the absolute sagittal measurements, we found differences in ethnicity for nasolabial angle, lower facial height (soft and hard tissue), total facial height, and mandible length (Table 2).

White patients exhibited significantly larger nasolabial angles (107.3 ± 11.9 degrees) compared to Latinx (101.2 ± 14.9 degrees, p = 0.04), Asian (93.8 ± 11.4 degrees, p < 0.001), and African American patients (82.5 ± 13.1 degrees, p < 0.001; Figure 2). African Americans had the largest osseous lower facial height of all racial groups with an average of 6.9 ± 0.7 cm, which was significantly larger than White (6.3 ± 0.6 cm, p = 0.006) and Asian (6.0 ± 0.5 cm, p < 0.001) patients. Similarly, soft tissue lower facial height was the largest in African American patients (7.4 ± 0.6 cm) and smallest in Asians (6.8 ± 0.5 cm, p = 0.03). Lastly, the mandible length was the smallest in Asian patients (7.2 ± 0.4 cm) and significantly different from the White patients (7.6 ± 0.5 cm, p = 0.01).

## Absolute coronal measurements

The only measurement made in coronal view was lateral mandibular flare. African Americans had the largest mandibular flare (0.4 ± 0.1 cm), whereas Latinx (0.2 ± 0.1 cm, p < 0.001) and Asian patients (0.2 ± 0.1 cm, p = 0.003) had the smallest (Figure 2).

## Facial ratios

Three of five ratios showed significant differences between the racial groups, including UFH/LFH, mandible width to total facial height, and zygoma width to total facial height (Table 3). UFH/LFH was the smallest in African American patients (0.8 ± 0.1) compared to White (0.9 ± 0.1, p < 0.001), Latinx (0.9 ± 0.1, p = 0.03), and Asian (1.0 ± 0.1, p < 0.001) patients (Figure 3). African American patients also had the smallest mandible width to total facial height ratio (0.8 ± 0.1) compared to White (0.8 ± 0.1, p = 0.03) and Asian (0.9 ± 0.1, p = 0.001) patients. However, Asian patients had the largest zygoma

**Table 2** Absolute lower facial measurements by race.

| Measurement parameter (mean ± SD) | Total (n = 204) | White (n = 123) | Latinx (n = 49) | African American (n = 12) | Asian (n = 20) | p value[a] |
|---|---|---|---|---|---|---|
| **Axial** | | | | | | |
| Zygoma width, cm | 13.1 ± 0.5 | 13.1 ± 0.5 | 13.2 ± 0.5 | 13.0 ± 0.4 | 13.5 ± 0.6 | **0.03** |
| Mandible width, cm | 10.0 ± 0.5 | 10.0 ± 0.5 | 9.9 ± 0.5 | 9.7 ± 0.6 | 10.1 ± 0.5 | 0.15 |
| **Sagittal** | | | | | | |
| *Nasal angles* | | | | | | |
| Nasolabial angle, ° | 102.8 ± 14.2 | 107.3 ± 11.9 | 101.2 ± 14.9 | 82.5 ± 13.1 | 93.8 ± 11.4 | **< 0.001** |
| Nasofacial angle, ° | 32.7 ± 5.0 | 32.9 ± 4.3 | 32.9 ± 4.1 | 32.1 ± 3.3 | 30.2 ± 3.7 | 0.15 |
| *Facial Height* | | | | | | |
| Upper facial height, cm | 5.7 ± 0.3 | 5.7 ± 0.3 | 5.6 ± 0.3 | 5.5 ± 0.4 | 5.6 ± 0.3 | 0.21 |
| Lower facial height, cm | 6.3 ± 0.6 | 6.3 ± 0.6 | 6.4 ± 0.5 | 6.9 ± 0.7 | 6.0 ± 0.5 | **< 0.001** |
| Total facial height, cm | 12.0 ± 0.8 | 12.0 ± 0.8 | 12.0 ± 0.7 | 12.5 ± 1.0 | 11.6 ± 0.6 | **0.03** |
| Soft tissue lower face height, cm | 7.1 ± 0.6 | 7.0 ± 0.6 | 7.3 ± 0.5 | 7.4 ± 0.6 | 6.8 ± 0.5 | **0.003** |
| *Mandible* | | | | | | |
| Mandible length, cm | 7.5 ± 0.5 | 7.6 ± 0.5 | 7.4 ± 0.6 | 7.6 ± 0.5 | 7.2 ± 0.4 | **0.01** |
| Ramus length, cm | 6.4 ± 0.5 | 6.4 ± 0.6 | 6.3 ± 0.6 | 6.3 ± 0.5 | 6.5 ± 0.3 | 0.60 |
| Mandibular angle, ° | 118.1 ± 6.4 | 118.4 ± 6.7 | 118.2 ± 6.3 | 118.3 ± 6.4 | 117.2 ± 4.9 | 0.80 |
| **Coronal** | | | | | | |
| Lateral mandibular flare, cm | 0.3 ± 0.1 | 0.3 ± 0.1 | 0.2 ± 0.1 | 0.4 ± 0.1 | 0.2 ± 0.1 | **< 0.001** |

Bolded values are statistically significant ($p < 0.05$).

[a] All p values are based on one-way ANOVA, except for zygoma width, nasofacial angle, and mandibular flare, which were evaluated using the Kruskal-Wallis tests.

**Table 3** Facial ratios by race.

| Measurement parameter (mean ± SD) | Total (n = 213) | White (n = 123) | Latinx (n = 49) | African American (n = 12) | Asian (n = 20) | p value |
|---|---|---|---|---|---|---|
| UFH/LFH | 0.9 ± 0.1 | 0.9 ± 0.1 | 0.9 ± 0.1 | 0.8 ± 0.1 | 1.0 ± 0.1 | **<0.001** |
| Mandible width to total facial height | 0.8 ± 0.1 | 0.8 ± 0.1 | 0.8 ± 0.1 | 0.8 ± 0.1 | 0.9 ± 0.1 | **0.005** |
| Zygoma width to total facial height | 1.1 ± 0.1 | 1.1 ± 0.1 | 1.1 ± 0.1 | 1.1 ± 0.1 | 1.2 ± 0.1 | **<0.001** |
| Zygoma width to mandible width | 1.3 ± 0.1 | 1.3 ± 0.1 | 1.3 ± 0.1 | 1.4 ± 0.1 | 1.3 ± 0.1 | 0.36 |
| Ramus height to mandible length | 0.9 ± 0.1 | 0.8 ± 0.1 | 0.9 ± 0.1 | 0.8 ± 0.1 | 0.9 ± 0.1 | 0.10 |

UFH/LFH, upper facial height/lower facial height.
Bolded values are statistically significant ($p < 0.05$).

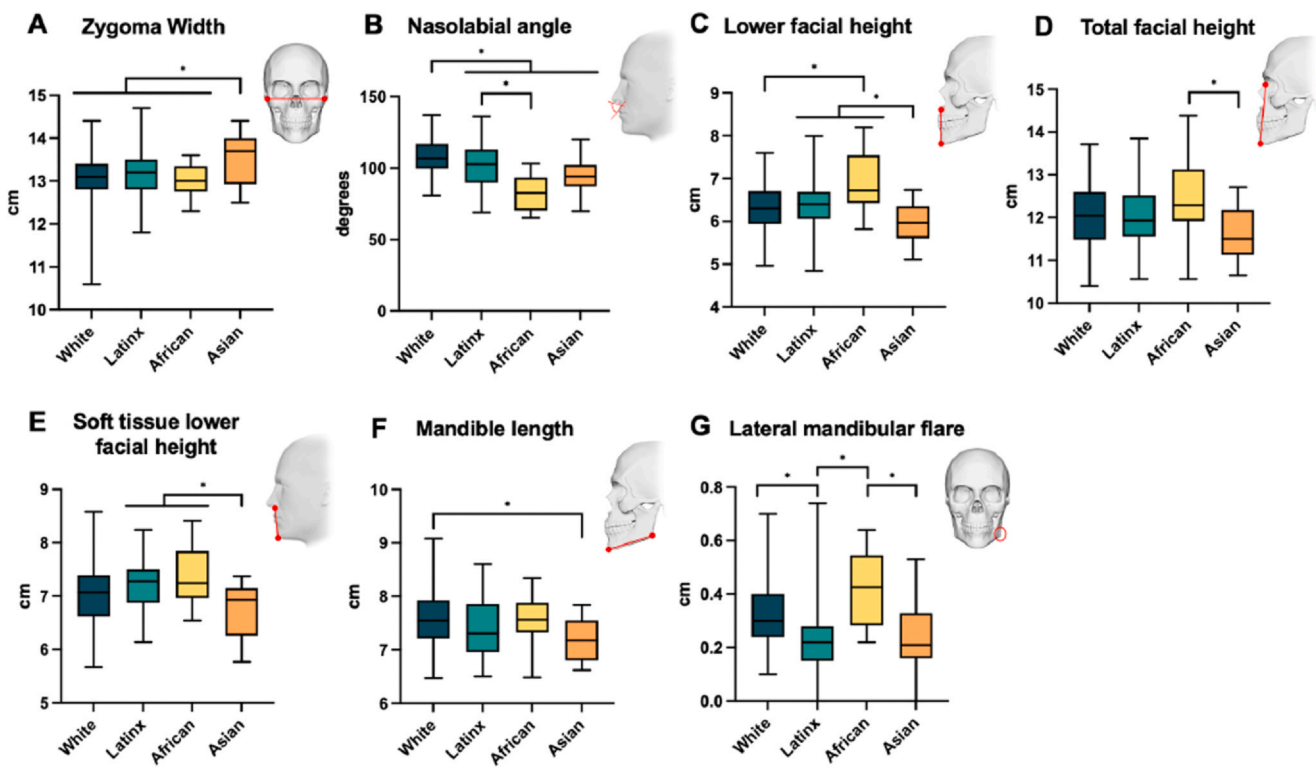


**Figure 2** Box and whisker plots of absolute lower face measurements with racial variation. (A) Zygoma width. (B) Nasolabial angle. (C) Lower facial height. (D) Total facial height. (E) Soft tissue lower face height. (F) Mandible length. (G) Lateral mandibular flare. *: Significantly different at $p < 0.05$.

width to total facial height ratio (1.2 ± 0.1) compared to all other patients (1.1 ± 0.1, $p < 0.001$; Figure 3).

## Race as a predictor in linear regression models

To account for the influence of other potential confounding variables such as age, BMI, and HRT duration in addition to race, linear regression models were constructed for all measurements that exhibited significant ANOVA comparisons. African American race was identified as a significant independent predictor for decreased nasolabial angle, increased lower facial height, decreased UHF/LFH ratio, and decreased mandible width to total facial height ratio (Table 4). Specifically, being African American was predicted to decrease the nasolabial angle by 17.26 degrees ($p = 0.004$) and increase the lower facial height by 0.58 cm ($p = 0.02$). Asian race was identified as a significant predictor for greater absolute zygoma width and greater ratio of zygoma width to facial height. Specifically, Asian race was a predictor for increase in zygoma width by 0.58 cm ($p = 0.008$). Additionally, Latinx race was identified as a significant predictor for decreased lateral mandibular flare (B=−0.12, $p = 0.02$). Conversely, mandible length, soft tissue lower facial height, and total facial height, despite showing significant ANOVAs, were not shown to include race as a significant predictor in the regression models. Furthermore, age was shown to predict decreased zygoma width (B=−0.02, $p < 0.001$), increased lower facial height (B=0.01, $p = 0.006$), and increased mandibular flare (B=0.002, $p = 0.04$), whereas BMI predicted increased zygoma width (B=0.03, $p < 0.001$) and increased lower facial height (B=0.02, $p = 0.002$).

## Discussion

Despite advancements in FFS surgical techniques, a knowledge gap exists in understanding the racial variations in

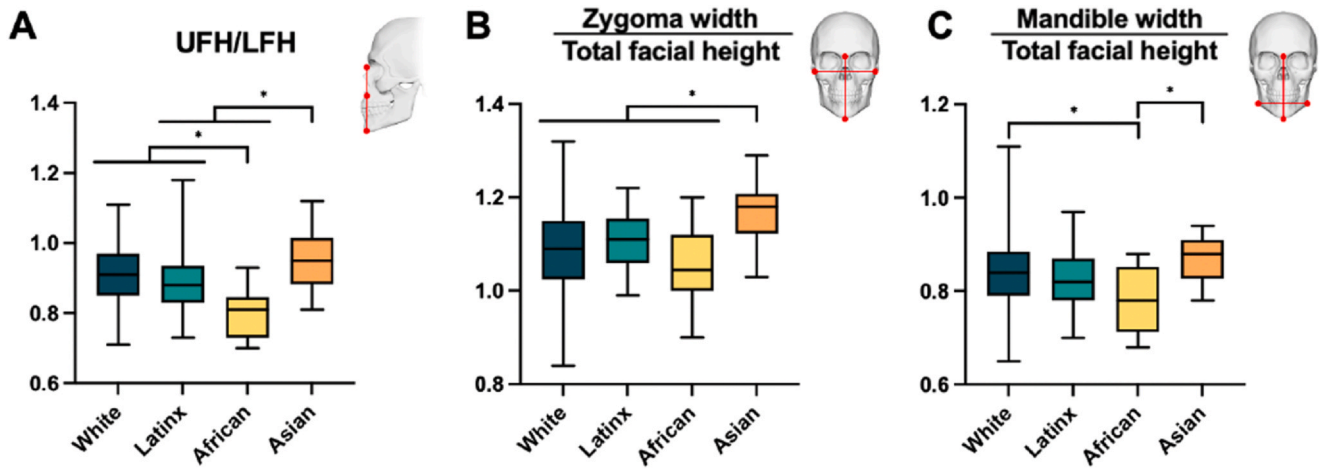


**Figure 3** Box and whisker plots of relative lower face proportions. (A) UFH/LFH. (B) Zygoma width/total facial height. (C) Mandible width/total facial height. *: Significantly different at $p < 0.05$.

craniofacial norms among TGNB individuals. Existing FFS literature predominantly focuses on White populations, and anthropometric studies mainly involve cisgender subjects.[5-8] This poses issues due to baseline physiologic variations and potential effects of long-term hormone therapy. The current study analyzes anthropometric differences in TGNB individuals in preparation for FFS. Our work demonstrated that: 1. African American groups are associated with decreased nasolabial angle and increased lower facial height, 2. Asian groups are associated with increased

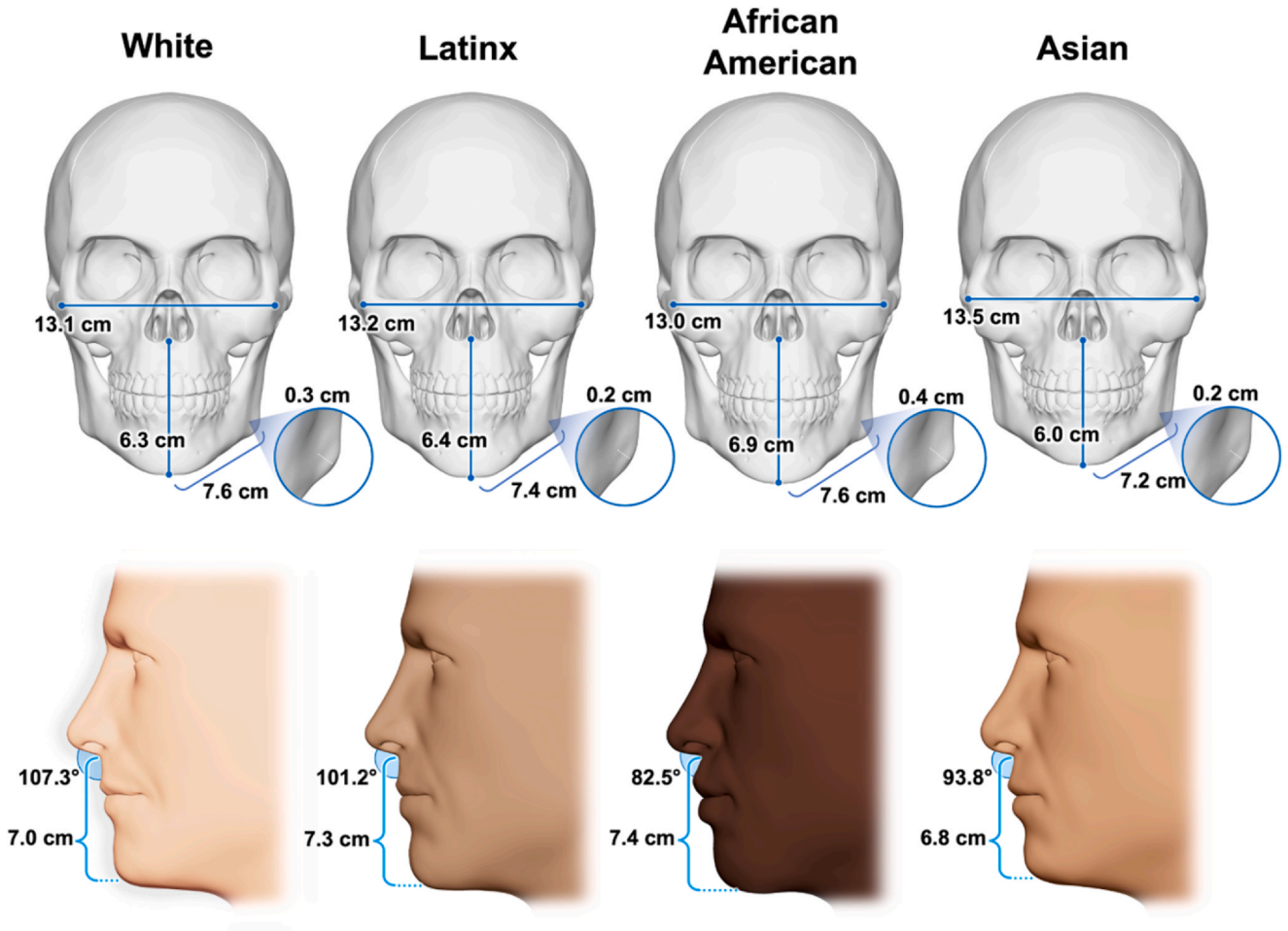


**Figure 4** Visual depiction of average lower face measurements by racial group. Significant differences include nasolabial angle, zygomatic width, lower facial height, soft tissue lower face height, total facial height, mandible length, and lateral mandibular flare.

zygoma width, and 3. Latinx groups are associated with decreased lateral mandibular flare. Careful consideration of these target areas may aid in addressing baseline ethnic variations operatively, as well as respecting patient preferences for preserving ethnic features.

## African American group differences

Our study found that African American race was a significant independent predictor of decreased nasolabial angle. Our findings align with the existing literature, which reports nasolabial angle as being smaller in cisgender African men (range 83.1–85.6 degrees) compared to White men (99.0–105.0 degrees).[3,9-11] This finding is also consistent with the cisgender female population, which shows that African women (80.8–90.6 degrees) tend to have smaller nasolabial angles compared to both White (102.9–119.2 degrees) and Asian women (92.9–102.5 degrees).[3,4,9,11-13] Traditionally, feminizing rhinoplasty aims to decrease overall nasal size while increasing the nasofrontal and nasolabial angles.[14-16] However, given that inter-racial variations are consistent across all genders, feminization rhinoplasty for TGNB individuals may not necessitate achieving a more obtuse nasolabial angle, as per the neoclassical standard.

Heir et al. discusses in great detail the variations in surgical techniques that may occur between three different types of African noses, as initially categorized by Ofodile et al.[17,18] For typical "African" noses with a wide and low nasal dorsum, bulbous tip, and short columella, a surgeon may consider greater dorsal augmentation, tip grafting, columellar grafting, and grafting to the nasal spine and maxilla. For "Afro-Caucasian" noses with higher and narrower bridges, less flared alae, and medially inclined nostrils, a surgeon may consider hump reduction, osteotomy and infracture, and a larger amount of alar cartilage excision. For "Afro-Indian" noses with a high and wide bridge, bulbous tip, and greater tip projection, a larger amount of alar cartilage excision and more extensive defatting relative to other nose types may be considered. Different rhinoplasty approaches have long been proposed for various ethnic groups regardless of gender.[19-22] Similarly, it is clinically valuable for gender-affirming surgeons to highlight with the patients during preoperative counseling on how to achieve feminization techniques without inadvertently conforming to the predetermined standards.

Our study also found that absolute and relative lower facial heights were consistently larger in African American patients. These findings align with previous studies that report increased absolute lower facial height in individuals of African descent compared to those of White and Asian individuals.[7,23-25] Studies with cisgender populations have consistently indicated that the lower face/total facial height ratio is a sexually dimorphic trait across races, with males typically exhibiting a greater relative lower face height.[26-28]

Traditionally, a feminine skull is characterized by a vertically shorter and more pointed chin.[29] However, it is crucial to carefully assess facial proportions to maintain overall balance and harmony of the face. Villavisanis et al. performed a study in which 1087 individuals rated 827 diverse facial photos on their level of masculinity-femininity and attractiveness.[30] Their findings revealed that among female photos, masculinity was associated with increased lower third facial height ratio. Meanwhile, among male photos, femininity was associated with increased middle third facial height ratio. Thus, it is especially pertinent in the TGNB population to focus on achieving balanced facial ratios as opposed to absolute measurements alone. Our findings suggest that African American patients may benefit from greater vertical chin reduction to achieve an ideal facial height ratio.

## Asian group differences

Our study found that Asian patients had the largest absolute and relative zygomatic width. This finding is consistent with that of Farkas et al., which found that zygomatic width was consistently greater in Singaporean Chinese, Vietnamese, Thai, and Japanese males compared to their North American White counterparts.[31] However, although absolute zygomatic width is consistently larger in Asian males compared to females, one study by Kramer et al. with 1697 White, 969 Black, and 488 Asian participants found that facial width to height ratio was sexually dimorphic for White and Black populations, but not for Asians.[32] Thus, while absolute measurements may vary, the proportional relationship between facial width and height is less likely to show significant differences between Asian males and females.

Most of the literature on FFS distinguishes between a flat zygoma in males and a prominent zygoma in females.[33] Although not universally indicated, feminization malarplasty may either involve reduction techniques to medialize the zygomatic complexes and reduce midface width, or placement of intraoral malar implants and fat grafting to increase soft tissue triangulation. Unlike most literature on Asian esthetics, which aims to reduce zygomatic width and prominence, FFS procedures more often simply augment the malar region for two reasons. The first reason is to maintain similar zygoma width to mandible width proportions to baseline, and the second reason is to minimize the risk of soft tissue ptosis from extensive lateral reduction of facial width.

## Latinx group differences

Our study found that Latinx race was a significant independent predictor for decreased lateral mandibular flare. FFS surgical approaches often aim to osteotomized or contour the mandibular angle for a softer appearance. Although TGNB patients may benefit from gonial contouring to reduce the lateral flare, it is possible that Latinx patients may require less extensive mandibular interventions. Although an angulated and square mandible is generally considered a more masculine face shape,[34,35] racial differences in mandibular contour are underreported. Studies on White or Asian patients favor a narrower mandible for femininity.[36,37] However, the preferences of Latinx and African American patients in esthetic surgery literature are notably lacking. Ethnically sensitive approaches in gender-

**Table 4** Predictors of facial measurements and ratios.

| Predictors | Nasolabial angle | | | Zygoma width | | | Lower facial height | | |
|---|---|---|---|---|---|---|---|---|---|
| | $R^2 = 0.23$, $F_{7194} = 8.07$, $p < 0.001$ | | | $R^2 = 0.16$, $F_{7199} = 5.43$, $p < 0.001$ | | | $R^2 = 0.19$, $F_{7194} = 6.63$, $p < 0.001$ | | |
| | B | 95%CI | p value | B | 95%CI | p value | B | 95%CI | p value |
| Age | −0.06 | −0.25, 0.13 | 0.52 | −0.02 | −0.02, −0.008 | **< 0.001** | 0.01 | 0.003, 0.02 | **0.006** |
| BMI | −0.35 | −0.69, 0.00 | 0.05 | 0.03 | 0.01, 0.04 | **< 0.001** | 0.02 | 0.009, 0.04 | **0.002** |
| HRT duration | −0.21 | −0.60, 0.19 | 0.31 | 0.02 | 0.001, 0.03 | **0.04** | −0.007 | −0.02, 0.01 | 0.41 |
| White race | 6.79 | −2.30, 15.87 | 0.14 | 0.20 | −0.16, 0.57 | 0.27 | −0.04 | −0.43, 0.34 | 0.83 |
| Latinx race | 1.82 | −7.76, 11.40 | 0.71 | 0.20 | −0.19, 0.58 | 0.31 | 0.04 | −0.37, 0.45 | 0.85 |
| African American race | −17.26 | −28.80, −5.72 | **0.004** | 0.09 | −0.37, 0.55 | 0.70 | 0.58 | 0.09, 1.07 | **0.02** |
| Asian race | −6.22 | −16.84, 4.39 | 0.25 | 0.58 | 0.15, 1.00 | **0.008** | −0.29 | −0.74, 0.16 | 0.20 |

| Predictors | Mandibular flare | | | Zygoma width to facial height ratio | | | UFH/LFH | | |
|---|---|---|---|---|---|---|---|---|---|
| | $R^2 = 0.18$, $F_{7195} = 6.19$, $p < 0.001$ | | | $R^2 = 0.17$, $F_{7194} = 5.83$, $p < 0.001$ | | | $R^2 = 0.19$, $F_{7194} = 6.63$, $p < 0.001$ | | |
| | B | 95%CI | p value | B | 95%CI | p value | B | 95%CI | p value |
| Age | 0.002 | 0.00, 0.004 | **0.04** | −0.002 | −0.004, −0.001 | **< 0.001** | −0.002 | −0.003, 0.00 | **0.01** |
| BMI | −0.001 | −0.004, 0.003 | 0.75 | 0.00 | −0.002, 0.002 | 0.72 | −0.003 | −0.005, −0.001 | **0.01** |
| HRT duration | 0.002 | −0.002, 0.006 | 0.38 | 0.002 | 0.00, 0.004 | 0.08 | 0.001 | −0.002, 0.003 | 0.48 |
| White race | −0.02 | −0.11, 0.08 | 0.75 | 0.01 | −0.04, 0.07 | 0.67 | 0.03 | −0.03, 0.08 | 0.38 |
| Latinx race | −0.12 | −0.21, −0.02 | **0.02** | 0.01 | −0.04, 0.07 | 0.68 | −0.003 | −0.06, 0.06 | 0.91 |
| African American race | 0.10 | −0.02, 0.21 | 0.10 | −0.04 | −0.11, 0.03 | 0.28 | −0.08 | −0.15, −0.007 | **0.03** |
| Asian race | −0.08 | −0.19, 0.02 | 0.13 | 0.07 | 0.008, 0.13 | **0.03** | 0.06 | −0.01, 0.12 | 0.10 |

| Predictors | Mandible width to facial height ratio | | |
|---|---|---|---|
| | $R^2 = 0.07$, $F_{7194} = 2.09$, $p = 0.047$ | | |
| | B | 95%CI | p value |
| Age | −0.001 | −0.002, 0.00 | 0.14 |
| BMI | 0.00 | −0.002, 0.002 | 0.81 |
| HRT duration | 0.001 | −0.001, 0.003 | 0.46 |
| White race | −0.005 | −0.05, 0.04 | 0.83 |
| Latinx race | −0.02 | −0.07, 0.03 | 0.51 |
| African American race | −0.06 | −0.12, 0.00 | **0.04** |
| Asian race | 0.02 | −0.04, 0.08 | 0.45 |

affirming surgery and shared decision-making can minimize the need for extensive interventions.

### Limitations

There are a number of limitations that are worth mentioning. First, although we included four major racial groups, the sample sizes for African American and Asian patients remain limited, which may affect the generalizability of results. This was partly accounted for by using nonparametric tests to adjust for limited sample size and lack of normality in some of the data. Additionally, any racial differences were verified in a multivariable logistic regression model that accounted for other confounding variables such as age, BMI, and HRT duration. Furthermore, patients who identified as Middle Eastern, Indigenous, or multiracial were excluded. Future iterations of this study will benefit from larger and more diverse sample sizes. This study also focused on preoperative characteristics of TGNB individuals, without long-term follow-up of patients after FFS. Although it may be of interest to include either a cisgender female or postoperative transfeminine comparison group, the goal of this study was to focus on TGNB patients with the purpose of surgical planning. By acknowledging baseline racial differences in facial proportions, craniofacial surgeons can have a better idea of what adjustments may be made operatively for each individual patient. Furthermore, the authors would like to acknowledge that race is a crude approximation for phenotypic variation and should be used with caution by clinicians without contributing to medical bias or stereotyping of which groups are considered to be more masculine or feminine in facial appearance. Preoperative counseling for FFS benefits from a balance of both cultural sensitivity and craniofacial esthetic techniques to relieve gender dysphoria in the diverse patient population.

## Conclusions

This study revealed notable racial variations in lower face measurements of TGNB patients, including nasolabial angle, zygomatic width, lower facial height, and lateral mandibular flare. These specific target areas should be carefully considered to account for possible baseline ethnic differences. Relative facial proportions may also be a more salient surgical planning tool for TGNB patients rather than absolute measurements alone to maintain overall harmony of the face.

## Ethical approval

Institutional review board approval was provided by the University of California, Los Angeles (IRB #22001299).

## Declaration of Competing Interest

All authors have no financial interests including products, devices, or drugs associated with this manuscript. JCL is a medical education consultant for Stryker. All sources of funds supporting the completion of this manuscript are under the auspices of the University of California, Los Angeles.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.bjps.2024.04.049.

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